

1. Document Overview

Purpose:

This document defines the functional requirements for an Order Management System (OMS). The OMS acts as a central orchestration layer for managing orders, inventory visibility, and integrations with downstream enterprise systems such as ERP, WMS, and Logistics platforms.

The document is intended for Business, Operations, Development, QA, and Architecture stakeholders.

In Scope:

- Order ingestion from multiple sales channels
- Order lifecycle management
- Order and inventory synchronization with external systems
- Operational monitoring, error handling, and manual fallback mechanisms

Out of Scope:

Functional Exclusions:

- Returns, refunds, and reverse logistics processes
- Customer account management
- Customer loyalty and promotions management

Document / Technical Exclusions:

- API specifications and interface contracts
- Database schema and physical data models
- UI design, layouts, and wireframes
- Infrastructure, deployment, and environment setup details

These items will be addressed in relevant design, UX, or technical specification documents where applicable.

Assumptions:

- OMS is not the system of record for payments
- Sensitive payment data (card number, CVV, PIN) is never stored in OMS
- External systems expose stable APIs for integration
- All sales channels provide a unique order identifier

2. Stakeholders

Business & Operational Users:

- **Operations Team**
Uses OMS for order monitoring, manual retries, exception handling, and fulfillment coordination.
- **Customer Support Team**
Uses OMS and commerce platform data to assist customers with order-related queries.
- **Finance Team**
Accesses order, payment, refund, and settlement data for reconciliation and audits.

Internal Technical Stakeholders:

- **Technical / Development Team**
Implements OMS integrations, workflows, validations, and system enhancements.
- **QA Team**
Validates order flows, integrations, failure scenarios, and regression cases.
- **DevOps Team**
Manages deployments, monitoring, alerts, retries, and system availability.
- **Security Team**
Ensures API security, data protection, access controls, and compliance with security standards.
- **Compliance Team**
Ensures adherence to regulatory requirements such as PCI compliance and audit logging.

External Systems & Vendors:

- **Commerce Platform**
Source system for customer-placed orders and customer-facing order updates.
- **Marketplace Platforms (Amazon, eBay, etc.)**
External sales channels providing marketplace orders to OMS.

- **Payment Gateway**
Handles payment authorization, capture, and transaction status updates.
- **ERP System**
Receives order and financial data for accounting and enterprise reporting.
- **Warehouse Management System (WMS)**
Handles inventory management and warehouse fulfillment activities.
- **Logistics / Courier Systems**
Provide shipment, tracking, and delivery status updates.
- **Notification Services (Email / SMS / Push)**
Send customer and operational notifications related to orders and failures.
- **Third-Party Integration Vendors / APIs**
Enable communication between OMS and external systems.

3. Role & Responsibility

Role	Responsibility
Business Owner	● Defines business objectives and success criteria ● Approves high-level scope and investment decisions
Product Owner	● Owns product vision and roadmap ● Prioritizes requirements and approves functional scope ● Acts as the primary decision-maker for functional trade-offs
Technical Business Analyst	● Elicits and analyzes business and system requirements ● Translates business needs into functional and integration requirements ● Defines end-to-end flows, data points, and integration contracts ● Prepares and maintains BRD, FRD, and supporting artifacts ● Validates requirements with business, Ops, and technical teams ● Supports QA during UAT and clarifies requirement-related queries

Operations Team	• Executes day-to-day OMS operations • Handles manual fallback processes during failures • Monitors order lifecycle, sync issues, and operational exceptions
Warehouse Team	• Manages inventory and fulfilment activities • Updates shipment and stock information in integrated systems
Logistics Partner	• Manages shipment, delivery, and tracking updates • Provides shipment status and proof of delivery information
Finance Team	• Oversees payment reconciliation and financial reporting • Handles payment disputes, refunds, and settlement processes
Development Team	• Designs and implements OMS functionalities and integrations • Fixes defects identified during testing and production
QA Team	• Creates and executes test scenarios based on requirements • Validates functional, integration, and regression test cases
DevOps Team	• Manages deployment, environment setup, and monitoring • Ensures system availability and performance post-deployment
Security & Compliance Team	• Validates security controls and compliance requirements (e.g., PCI) • Reviews access controls, audit logs, and data protection measures
Customer Support Team	• Handles customer queries related to orders, payments, and delivery • Coordinates with Ops and technical teams for issue resolution
Third Party / Integration Vendors	• Provide and maintain APIs for payments, logistics, notifications, etc. • Provide and maintain APIs for

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4. Functional Requirements - Order Management

FR-01: Order Ingestion from Sales Channels:

The OMS shall ingest orders from configured sales channels such as Website, Mobile App, and Marketplaces.

Acceptance Criteria:

- **AC-01.1:** Website orders shall be available in OMS within 10 minutes of successful order placement.
- **AC-01.2:** Orders from non-website channels shall be synchronized based on a configurable batch interval.
- **AC-01.3:** Each order shall contain minimum mandatory attributes: order number, sales channel, order status, order date/time, item details, totals, payment reference, and shipment reference (if available).
- **AC-01.4:** OMS shall validate order uniqueness using order number and sales channel before persistence.

FR-02: Order Status Initialization and Management:

The OMS shall assign and manage order statuses based on configured business rules.

Acceptance Criteria:

- **AC-02.1:** Initial order status shall be mapped based on payment and validation outcomes.
- **AC-02.2:** Order status transitions shall follow configured state transition rules.
- **AC-02.3:** Unauthorized status updates shall be blocked based on role-based access control.

FR-03: Role-Based Order Visibility and Edit Permissions:

The OMS shall provide role-based access for order visibility and modifications.

Acceptance Criteria:

- **AC-03.1:** Order fields editable in OMS shall be restricted based on configured user roles.
- **AC-03.2:** All manual edits shall be audit logged with user, timestamp, and field-level changes.

FR-04: Order De-duplication:

The system shall validate the uniqueness of each order before processing to prevent duplicate records.

Acceptance Criteria:

- **AC-04.1:** Duplicate order numbers are rejected or flagged.
- **AC-04.2:** Duplicate orders are not persisted in the system.
- **AC-04.3:** Validation occurs prior to further processing or synchronization.

5. Functional Requirements - External System Integrations

FR-05: Order Data Sync to External Systems

The OMS shall synchronize order data with configured external systems such as ERP, WMS, and Logistics systems.

Acceptance Criteria:

- **AC-05.1:** Integration configurations shall be managed via admin panel (enable/disable, API details, retry count, sync interval).
- **AC-05.2:** OMS shall log sync execution details including sync time, records processed, status, and response.
- **AC-05.3:** Duplicate order transmission shall be prevented using order ID and target system validation.

FR-06: Inventory Synchronization

The OMS shall receive inventory updates from ERP/WMS and publish inventory availability to sales channels.

Acceptance Criteria:

- **AC-06.1:** Inventory sync shall support batch processing with configurable intervals.
- **AC-06.2:** Inventory discrepancies shall be logged and flagged for operational review.

6. Functional Requirements – Error Handling & Recovery

FR-07: Automated Retry Mechanism

The OMS shall retry failed operations for critical workflows based on configurable retry policies.

Acceptance Criteria:

- **AC-07.1:** Retry count and retry interval shall be configurable per integration.
- **AC-07.2:** Failures exceeding maximum retry attempts shall trigger alerts.

FR-08: Manual Fallback Mechanism

The OMS shall support manual execution of failed processes by authorized operational users.

Acceptance Criteria:

- **AC-08.1:** Ops users shall be able to trigger single or bulk reprocessing of failed orders.
- **AC-08.2:** Manual actions shall be audit logged.

FR-09: Notification Management

The OMS shall notify operational stakeholders on critical failures.

Acceptance Criteria:

- **AC-09.1:** Email notifications shall be sent when order or integration sync fails after maximum retries.
- **AC-09.2:** Notification recipients shall be configurable.

7. Non-Functional Requirements

Performance

- Customer order confirmation notification (email/SMS) should be sent within ≤ 60 seconds after successful order placement confirmation in the commerce platform.
- Order data sync from Marketplace and Commerce Platform to OMS should run in a batch process with a maximum interval of 10 minutes.
- During peak campaigns and sale days, the system should support 3-5 x more orders than normal volume per day without degradation in order placement response time or sync SLAs
- Order and inventory data sync from OMS to external systems (ERP, WMS, Logistics) should occur via batch processing with a maximum interval of 10 minutes.
- System performance metrics (order placement latency, sync duration, failure rates) should be monitored and accessible to Ops and Tech teams.

Scalability

- The OMS should support the onboarding of additional sales channels without requiring major redesign or architectural changes.
- During peak campaigns and sale events, the system should scale to support 3–5x the normal daily order volume.

- Integration with additional warehouses or fulfillment partners should work under the existing OMS system design.

Security

- All payment-related operations must comply with PCI-DSS standards
- Sensitive API data, including customer, order, and payment-related information, must be secured and encrypted to prevent data tampering and unauthorized access.
- All APIs must be authenticated and authorized, with role-based access controls where applicable.
- The system must not store sensitive payment information such as card number, CVV, or PIN to prevent payment data leakage

Reliability & Availability

- Configurable retry mechanisms should be implemented for critical operational activities such as order sync, inventory updates, and shipment updates.
- Manual fallback processes should be available for critical operations where automation fails due to scenarios such as reaching maximum retry limits, API timeouts, or data validation errors, especially in Phase 1.
- The OMS should gracefully handle failures or delays in integrations with external systems (ERP, WMS, Logistics) without blocking core order management operations.
- Critical operational failures should be visible through alerts or dashboards to ensure timely detection and resolution.

There are a few more details available in a separate excel document which cover the below topics:

- **Data points & integration contracts**
This section covers the integrated system details with key data information and business purpose.
- **Risk scenarios and failure handling**
This section covers possible risk scenarios, business and customer impact, the detection ways and the failure handling processes.